Sean Luka Girgis

Data Platform Engineer | Python, SQL, Data Quality, Ingestion & Production Reliability

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Professional Summary

Senior Data Platform Engineer with 20+ years of enterprise technology experience across Python, SQL, data ingestion, ETL/ELT pipelines, data quality, observability, schema design, production support, financial-services technology, and stakeholder-facing analytics. Strong background building reliable data workflows, validating pipeline outputs, investigating source data, documenting data mappings, troubleshooting production issues, and converting operational data into trusted reporting and decision-support datasets. Best aligned to hands-on Python/SQL data platform roles requiring API/file ingestion, data quality checks, freshness/completeness/correctness validation, pipeline reliability, schema ownership, production support, cross-functional collaboration, and comfort operating with ambiguous or poorly documented source systems.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas/PySpark data pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational feeds into curated datasets for reporting, analytics, forecasting, validation, and decision support.
- Designed AWS and database-backed data platform components using S3, AWS Glue, Athena, Redshift, Oracle, SQL, and PySpark to support scalable ingestion, transformation, querying, and downstream consumption.
- Investigated source data behavior across monitoring platforms, infrastructure systems, Oracle datasets, and AWS-backed reporting layers to determine clean mappings into trusted reporting models.
- Implemented data quality and reconciliation practices across BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, and AWS-backed reporting layers to improve freshness, completeness, consistency, and correctness of pipeline outputs.
- Owned schema-aware reporting datasets from ingestion through production support, including metric definitions, historical retention patterns, validation checkpoints, and downstream stakeholder usage.
- Troubleshot production data issues by tracing source feeds, transformation logic, SQL behavior, missing/late data, failed outputs, latency symptoms, and reporting discrepancies.
- Prepared ML-ready datasets for Prophet and scikit-learn forecasting workflows, supporting capacity planning, bottleneck prediction, trend analysis, and business decision support.
- Partnered with infrastructure, analytics, application, engineering, and business teams to translate ambiguous data questions into reliable datasets, dashboards, documentation, and operational recommendations.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Built analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.

- Led the FAST project, investigating real-user and synthetic monitoring source data to identify data quality issues, performance trends, and optimization opportunities.
- Created dashboards and reporting views that converted raw monitoring data into actionable performance, reliability, and stakeholder insights.
- Partnered with engineering teams to explain tradeoffs, validate findings, and support practical performance improvements.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM/Introscope SME for a large financial-services environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable data pipelines.
- Designed dashboards, alert rules, SLA thresholds, reporting standards, and technical documentation used by application, middleware, infrastructure, and operations teams.
- Provided troubleshooting guidance, onboarding support, and knowledge sharing to improve adoption of monitoring standards and operational playbooks.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom applications under load to identify throughput limits, JDBC bottlenecks, thread contention, heap pressure, CPU constraints, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, regression thresholds, test findings, and release-readiness notes used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Automated schema conversion, data validation, and phased cutover support for a mission-critical enterprise migration.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.
- Supported technical design documentation and integration patterns for structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported high-availability C++/Pro*C/PL/SQL billing interfaces and data feed processes for telecom-scale transaction systems.
- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, and production housekeeping.
- Implemented data-feed business logic using C/C++, Pro*C, PL/SQL, SQL scripts, XML, shell automation, and enterprise messaging interfaces.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, XML, JSON, Perl, KornShell/ksh, C++, C, Java

Flagship Projects

Serverless Lakehouse Platform (AWS)
Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, transformation, curated analytics datasets, query optimization, reporting access, and validation checkpoints.
Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

HorizonScale — Capacity Forecasting Engine
Built Python/PySpark forecasting workflows to process infrastructure telemetry, prepare forecasting datasets, validate pipeline outputs, and deliver capacity and bottleneck insights through dashboards.
Technologies: Python, Pandas, PySpark, SQL, Prophet, scikit-learn, Data Validation

AI-Powered Job Search Pipeline
Built a Python automation pipeline with semantic duplicate detection, LLM-based scoring, structured JSON artifact generation, quality gates, document rendering, and application tracking.
Technologies: Python, JSON, LLM API, FAISS, Sentence Transformers, Quality Gates

Enterprise APM Reporting Automation
Built Perl and KornShell/ksh automation for extracting, transforming, validating, and distributing performance telemetry across large CA APM enterprise monitoring environments.
Technologies: KornShell/ksh, Perl, Unix/Linux, CA APM, SQL, Monitoring, Automation